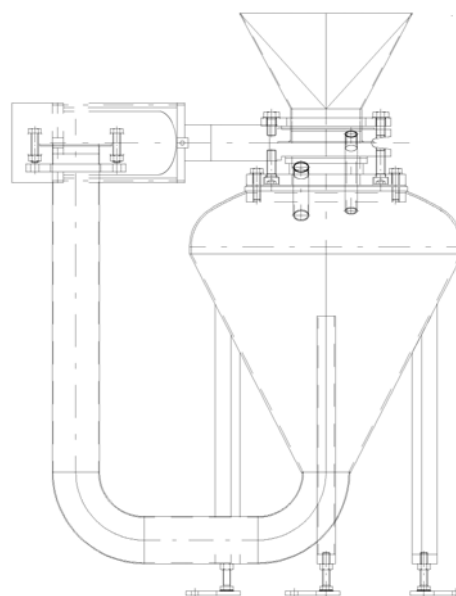
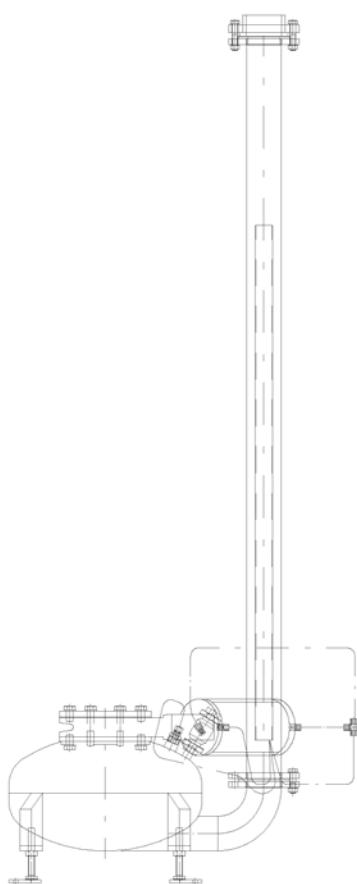


User- and Maintenance manual



0120-3

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Foreword

Meyn Food Processing Technology B.V. is one of the worlds leading designers and manufactures of systems and equipment for the poultry processing industry. Meyn's flexible service and support organisation and its constant emphasis on product innovation through extensive research and development, enable Meyn to supply today's customers with the solutions of tomorrow.

Regardless of the size or complexity of the process, Meyn will supply its customers with situation specific solutions that are critical in reaching their objectives. Therefore, Meyn combines a solid knowledge of the customers business with expert process design skills. This is crucial, as only well focused solutions generate the desired returns.

Our ambition is to supply solutions that support fully controlled in line processing; handling and tracking every product or batch throughout the entire facility while controlling the quality of the output, combined with the ability to match the incoming birds with the sales orders in the most optimal manner.

Meyn is worldwide represented by a network of subsidiaries and agents. In order to supply complete integrated solutions to our customers, Meyn has a number of strategic partners in the following processing sectors:

- Bonescan.
- Mechanical deboning.
- Logistic systems.
- Refrigeration.
- Water treatment.
- Vacuum transport.
- Rendering.
- Convenience food.

This manual

This manual is intended for the operator(s) of this machine. It's a means for information and instruction, so you can use the machine correctly and safely.

The manual consists of 6 chapters and a few appendices:

- Chapter:1. Introduction, includes general information about the machine and the supplier.
- Chapter:2. Safety, includes a description of the most important safety aspects associated with the use of the machine.
- Chapter:3. Installation and set-up, includes a description of the actions that must be performed before the machine can be used.
- Chapter:4. Cleaning, includes a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.
- Chapter:5. Maintenance, includes a description of the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation.
- Chapter:6. Disassembly/assembly and adjustments describes the disassembly and assembly of parts needed by reinstallation of the machine to another place or environment and to ensure that the machine performs safely and efficiently.
- Appendix:. Appendices, includes aspects/details for the machine.

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1 Introduction

This chapter contains technical data about the machine and gives a description of the goal and working of the machine.

1.1 Technical specifications

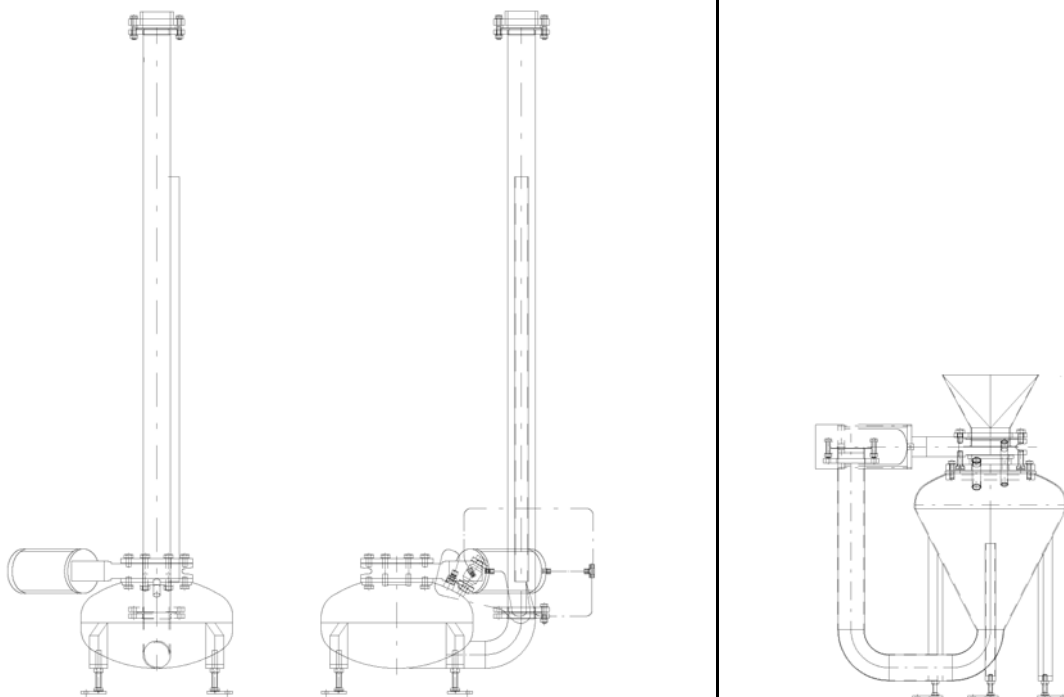


Figure 1.1 Sizes dry offal blow vessel.

Specifications:			
Product	** Intestines, Feather, Crushed heads?	Diameter output pipe	** 110, 156 mm
Capacity	**9,000 8,000	volume	** 60, 160 ltr
Diameter supply pipe	**100 150 300mm		

Table 1.1 Technical specifications.

1.2 General arrangement drawing

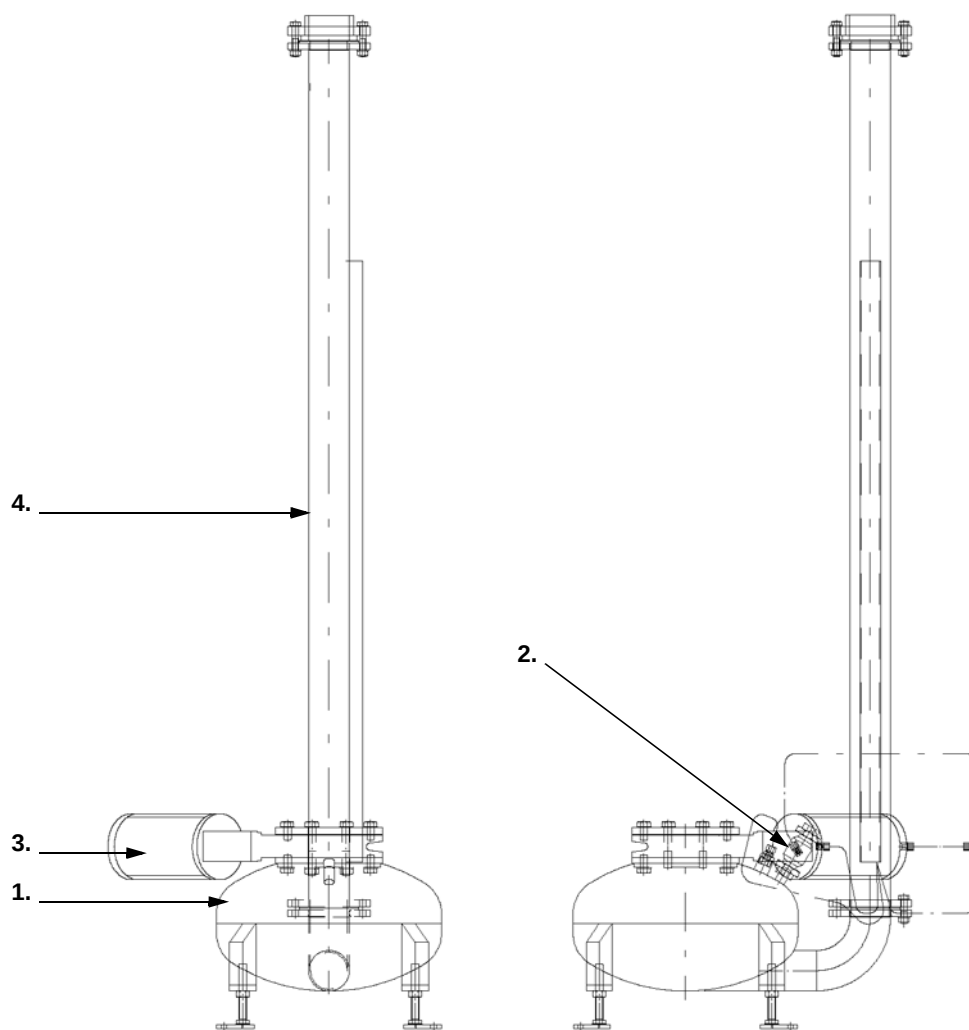


Figure 1.2 dry offal blow vessel.

The Dry offal blow vessel consists of the following main parts:

1. Tank
2. Valve
3. Knife valve
4. Pipes

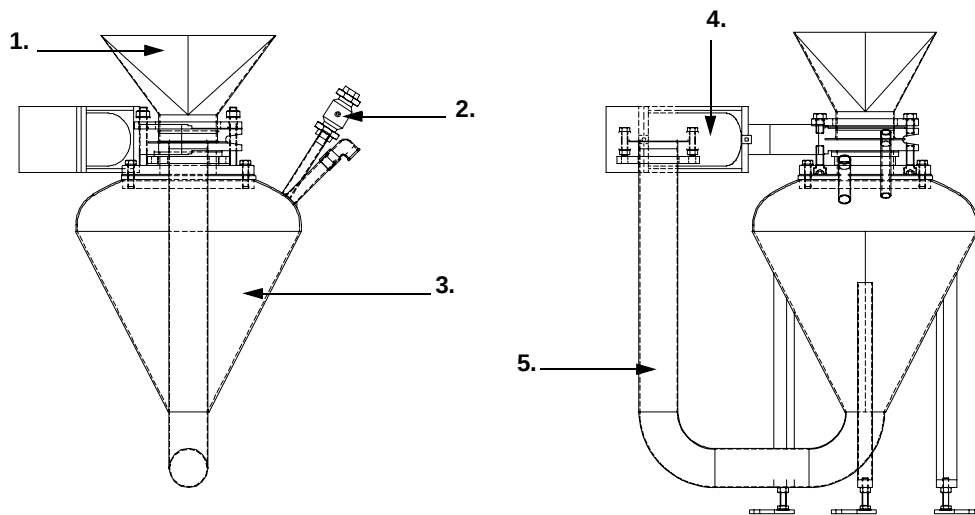


Figure 1.3 dry offal blow vessel.

The Dry offal blow vessel consists of the following main parts:

1. Infeed funnel
2. Bush valve
3. Tank
4. Knife valve
5. Pipes

2 Safety

This chapter contains a description of safety aspects relating to the machine. It is particularly important that you are fully aware of the contents of this section. The Meyn Food Processing Technology B.V. has taken all possible steps to provide you with all the necessary information concerning any dangers that may exist when the machine is used. The customer is responsible for understanding and applying these safety rules.



If any of these aspects are not clear, please ask the manufacturer for an explanation.

2.1 The most important safety risks

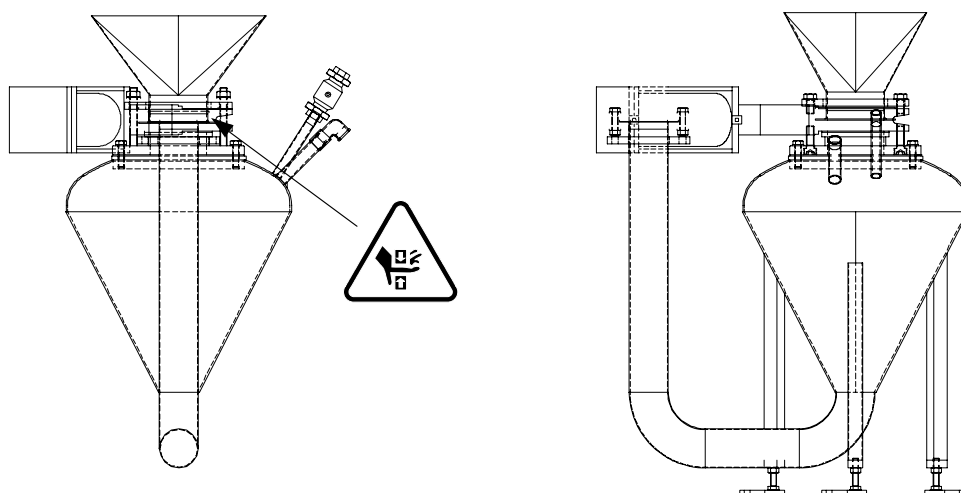


Figure 2.4 Safety risks.

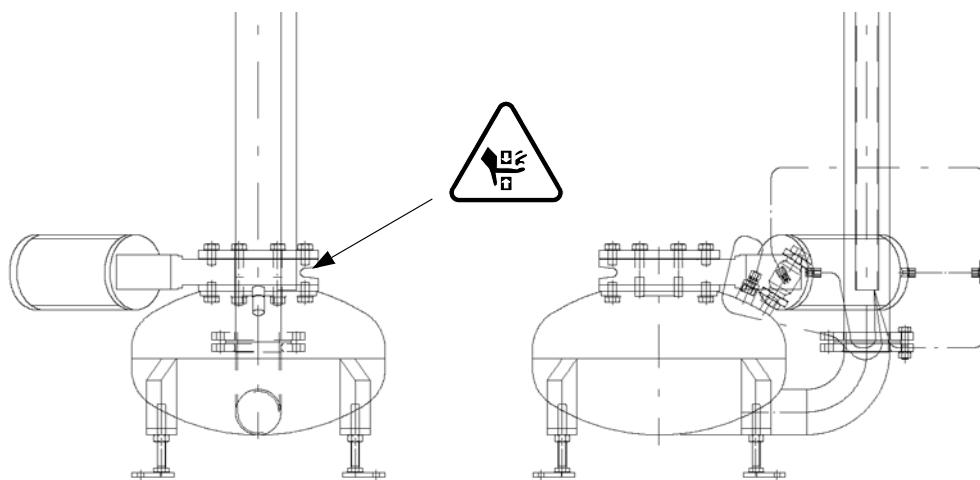


Figure 2.5 Safety risks.

- Do not put your hands or other parts of your body in the funnel. The valve may cause injury.

2.2 Safety measures to be taken

Before and during use of the machine it is important to take a number of safety measures and instructions into account, which are described in the sections below.

2.2.1 Preparing for safe operation

For safe operation of the machine and preceding any act, it is essential that a number of safety measures will be taken: these are mentioned in the following paragraph.

- First read the concerning instructions given in this manual, before you perform any work on the machine.
- Wear footwear which prevents slipping; mind loose-hanging clothing.
- You must take note of all the safety symbols attached to the machine.
- Check that the power supply complies with the locally applicable regulations.
- Ensure that the machine is correctly placed in a stable position.
- Ensure that the operation, minor maintenance activities and activities concerned with installing or moving the machine are only performed by qualified personnel, that is to say by personnel who have the necessary skills and are familiar with the contents of this manual.
- Modifying the machine in any way, could endanger the safety and proper functioning of the machine.

2.2.2 Safety measures during operation

For safe operation of the machine, it is essential that a number of safety instructions are taken during an act: these are mentioned in the following paragraph.

- Ensure that no persons are on, in or under the machine while it is in use.
- Keep the area around the machine free from materials and other obstacles.
- Always stop the machine if a product becomes jammed.
- Never put your hands or any other part of your body close to rotating parts while the machine is running.
- Mind to place warning notices with the control panel of the machine while performing maintenance operations to the machine.
- Never put your hand or other parts of your body, object or tools in the infeed funnel.

2.3 Know the pictograms

Machines manufactured by Meyn Food Processing Technology B.V. are provided with safety and protective devices. Even so, it is important to be careful when you are performing any operation on the machine. The following pictograms indicate possible dangers. The pictograms are applied to the machine and are also printed in this handbook at those places where risky activities are described.



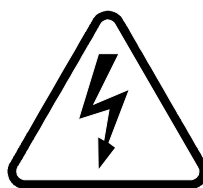
Check regularly to ensure that all the pictograms are still in the correct positions on the machine. If any pictogram has become detached or unreadable, apply a new one immediately.



Danger
Moving parts



Danger



Danger
High Voltage

3 Installation

This chapter gives a description of the actions that must be performed before the machine can be used.

3.1 Installing the dry offal blow vessel

Place the dry offal blow vessel level on a level and stable floor.

When the machine is delivered, the adjustment of the movable parts of the machine has already been done at the factory. In chapter 7, Disassembly/assembly and adjustments, is given the information about disassembly and assembly of parts you must do to ensure that the machine performs optimal.

3.1.1 Required floor area

Floor area required: The surface area (L*W) occupied by the machine is given in paragraph 1.1 "Technical specifications". Nevertheless, it is desirable to install the machine on a floor area that makes it easy to clean and maintain. Meyn advises you to keep a clear space of 1 meter on all sides of the machine.

3.2 The connection of the dry offal blow vessel

Connect all pipes.

4 Cleaning

This chapter gives a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.

4.1 Cleaning preparations

- During cleaning you are advised to wear detergent- and disinfectant proof clothing. Protect your eyes and skin.
- Ventilate the room in which you are working.
- After production, all products should be removed from the room in which the machine is installed. Remove coarse dirt.



Switch off the main switch of the machine before cleaning/adjustments.

4.2 General cleaning schedule

<i>During breaks</i>	<i>Daily after production</i>	<i>Monthly *</i>
High-pressure water cleaning.	High-pressure water cleaning. Alkaline foam cleaning. High-pressure water cleaning. Disinfection. High-pressure water cleaning.	Acid foam cleaning.

Table 5.1 The cleaning schedule of the machine.

*) or as often as necessary.

- Clean the machine from top to bottom.
- During cleaning, pay extra attention to:
 - The inside of the dry offal blow vessel.
- During cleaning, never point the high-pressure water jet to the control panel or motor.
- For proper functioning, after cleaning the moving parts of the machine should be sprayed with a vegetable oil.

4.2.1 High-pressure water cleaning

With high-pressure water cleaning nearly all visible dirt will be sprayed off the machine. This cleaning is a preparation for the daily foam cleaning and disinfection. Furthermore, a high-pressure water jet can be used for localized cleaning during production breaks.

- High-pressure water cleaning should always be performed as close to the machine as possible. Stop the machine before starting the high-pressure water cleaning
- Prevent people from getting in the way of the high-pressure jet. Always hold the spray gun with both hands.

For high-pressure cleaning use the following equipment:

- A fixed piping system, delivering water-pressure with a minimum of 3,000 kPa and a maximum of 6,000 kPa and with a water temperature of approx. 55°C. at the spray-mouth
- A spray lance with a downward bent nozzle. Such a spray lance has the advantage of not transporting the dirt upwards
- A spray nozzle of the flat-jet type with a spray-angle of 15° to 25° and a water consumption of 800 to 1000 litres/hour at a water pressure of 6,000 kPa

A water temperature of 55°C is advised, since a higher temperature may cause sticking of proteins and with a lower temperature the fat will not be removed properly. The pressure should not be too high, to avoid unnecessary mist, and to protect e.g. bearings.

The unit kPa (kilopascal) is SI standard for pressure:

<i>New</i>	<i>Old</i>
100 kPa	1 bar
3000 kPa	30 bar
10000 kPa	100 bar

Table 5.2 **SI.**

4.2.2 Alkaline foam cleaning

Alkaline foam cleaning ensures that fat, proteins etc. will be degraded chemically.

Foam the machine from top to bottom. Foaming may be performed while the machine is running slowly, provided sufficient distance is kept to the moving parts of the machine. Mind the other machines.

After foam cleaning remove the dirt and foam residue thoroughly by means of the high-pressure water jet.

For alkaline foam cleaning use the following equipment:

A fixed piping system for foam delivery.

A large spray-nozzle of the flat-jet type, through which the foam can be distributed equally along the surface.

The alkaline foam detergent used should not affect the materials which it will contact. Consult your detergent supplier to know which detergents will fulfil these specific requirements and which cleaning directions to follow.

Obey these instructions. Incorrect application may be dangerous

Please avoid:

A too high concentration.

A too long contact

Not rinsing

Mixing with other products

It is advisable not to use a combination of detergents and disinfectants ('detergent sanitizers'). This leaves the machine too dirty to be disinfected effectively.

A combination of these two methods is of no use at all.

4.2.3 Disinfection

Alkaline foam cleaning alone does not lead to a complete killing and removal of bacteria, fungi, viruses, algae and yeasts. For that purpose a separate disinfectant is needed. Disinfect the machine daily, after the alkaline foam cleaning.

Disinfection only makes sense when the surfaces are visibly clean after the alkaline foam cleaning. Spray the disinfectant onto the machine and surrounding area. When disinfecting, it is difficult to judge just how thorough you have been.

We advise you to disinfect the machine and the area immediately surrounding it in an established pattern.

After applying, the remaining foam rests should be removed by means of high-pressure water jet.



Damage may be caused by not thoroughly rinsing the disinfectant. Evaporation of chlorous drops will lead to a very high concentration of chlorine. In combination with long contact-times can cause corrosion to metals.

For disinfection use the following equipment:

A fixed piping system.

A spray lance with downward-bent top.

A spray nozzle of the flat-jet type, with a spray angle of 15° to 25°.

- The used disinfectant should not affect the materials which it will contact. Consult your detergent supplier to know which detergents will fulfill these specific requirements and which cleaning directions to follow. Obey these instructions. Incorrect application may be dangerous.

4.2.4 Acid foam cleaning

"Hard" water causes calcium and rust deposits on the machine. In order to remove these, a regularly performed acid foam cleaning is necessary. Depending on the hardness of the water this may be either once a month or more often if necessary.

Foam the machine from top to bottom. Foaming may be performed while the machine is running, provided sufficient distance is kept from the moving parts of the machine. Mind the other machines and the overhead conveyor. After affecting, the dirt and foam residue should be removed thoroughly by means of the high-pressure water jet.



Contact between acids and chlorous products results in a poisonous gas. Ensure that these products do not come in contact with each other. Ventilate the room in which you are working.

For acid foam cleaning use the following equipment:

A fixed piping system for foam supplying.

A large spray-nozzle of the flat-jet type, that distributes the foam equally along the surface.

The used acid foam detergent should not affect the materials with which it comes into contact. Consult your detergent supplier which detergents fulfil these specific conditions and the concerning directions for operation. Obey these instructions carefully. An incorrect application may be dangerous.

It is advisable not to use spring water for the production of foam; this water may contain a high degree of chloride. Tap water reduces the chance of rust.

4.3 Advising cleaning chemicals to the customer.

Customers sometimes ask Meyn Food Processing Technology B.V. to advise them about cleaning chemicals and disinfection chemicals. As a result of these questions some recommendations are as follows as stated below.

We recommend:

- To contact the major manufacturers or suppliers of these chemicals, like Henkel, Diversey etc. These companies have a good knowledge of cleaning chemicals and their corrosive or destructive properties on materials.
- Do not buy unknown (cheap) products.
- To follow all user instructions carefully.
- Meyn Food Processing Technology does not recommend to use specific cleaning chemicals, because we do not know which chemicals are available in each specific country. We can not know the (secret, complex) composition of the products. Besides this, Meyn Food Processing Technology does not accept any responsibility for any damage caused by wrong application of cleaning chemicals.
- The above mentioned can only be avoided by a direct contact between user and supplier of cleaning chemicals.

5 Maintenance and trouble shooting table

This chapter describes the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation. We advise you to maintain the machine regularly. This lengthens the working life of the machine and reduces the chance of failures.

5.1 Lubricants

Lubricants used in the food processing plants are categorized and registered by the NSF in two groups:

- H1 lubricants, are used when incidental contact with food products cannot be excluded.
- H2 lubricants, are used for all other applications in a food plant.

For all bearings and moving parts Meyn advises to use a multi purpose NLGI 2 H1 grease with the following characteristics;

- Excellent water resistance.
- Chemical resistant.
- Food grade H1.
- High pressure (EP) properties.
- Compatible with other greases.

It is advised not to mix greases!

Meyn is using "Jax Halo Guard FG2" as a general purpose grease for most applications. Other greases are used for:

- For high speed rotations (> 5000 rpm).
- For low temperatures (< - 5°C or 23 F).
- For gearboxes.
- For Hydraulics.

For the hydraulic height adjustments of ECP evisceration machines Hydraulic Fluid 32 Food grade H1 is used.

Meyn is using "Rocol Foodlube multi paste" in order to mount threaded stainless steel parts like bolts and nuts together, a special paste is used that prevents seizing of the threaded parts and allows for disassembly when needed.

5.2 Checklist - daily

<i>Part</i>	<i>Check</i>
Valve	Check the functioning of the bush valve. Replace the bush valve if necessary.
Knife valve	Check the functioning of the knife valve. Replace the knife valve if necessary.

Table 6.1 Checklist - daily.

6 Disassembly/assembly and adjustments

This chapter describes how to disassemble, assemble and adjust components of the machine after maintenance and/or replacement of parts.

6.1 Basic adjustments

The basic adjustments to the various component parts of the units in view of each other are carried out in the factory. However, as a result of the adjustments to the parts and for local circumstances, it may be necessary to re-insert, and or readjust the basic adjustments.

Contact service department of Meyn Food Processing Technology B.V. before readjust main parts of the machine.

